If the desk were built from the beginning

The same jobs, the same research rules, rebuilt around what the 2026-10 review found. Every design choice below points at the failure it prevents.

How the desk runs today is in ARCHITECTURE.md. The rules each program must keep are in screeners/README.md and qp/README.md.

What the desk actually does

TradingView is where the candidates come from. It is **not** where most of the desk's work happens.

Job	What it is	Weight
1. Discovery	each tool's screeners ask TradingView's scanner for candidates	one request per screener per scan
2. The card	every candidate is filled with everything known about it: derived numbers, short interest, borrow, news and catalyst, market context, industry and group, the full CANSLIM reading, themes, score, shortlist marks	**the biggest part of the desk**: ten external sources, about twenty steps, each with its own cache, freshness and fallback
3. The market	the Market tab: the desk's live read (indexes, regime, sectors, hot sectors) and O'Neil's method (the market model M, industry groups L), also stamped on every card	every scan, plus nightly
4. Research	freeze the photo (R1), measure what moved (R3), join them (R4), judge screeners, train the score	daily
5. Charts, strategies, backtests	qp, on the stocks the tools found	on demand
6. The journal	every trade after the fact	on demand, plus broker sync

The design has to be built around job 2, not around job 1.

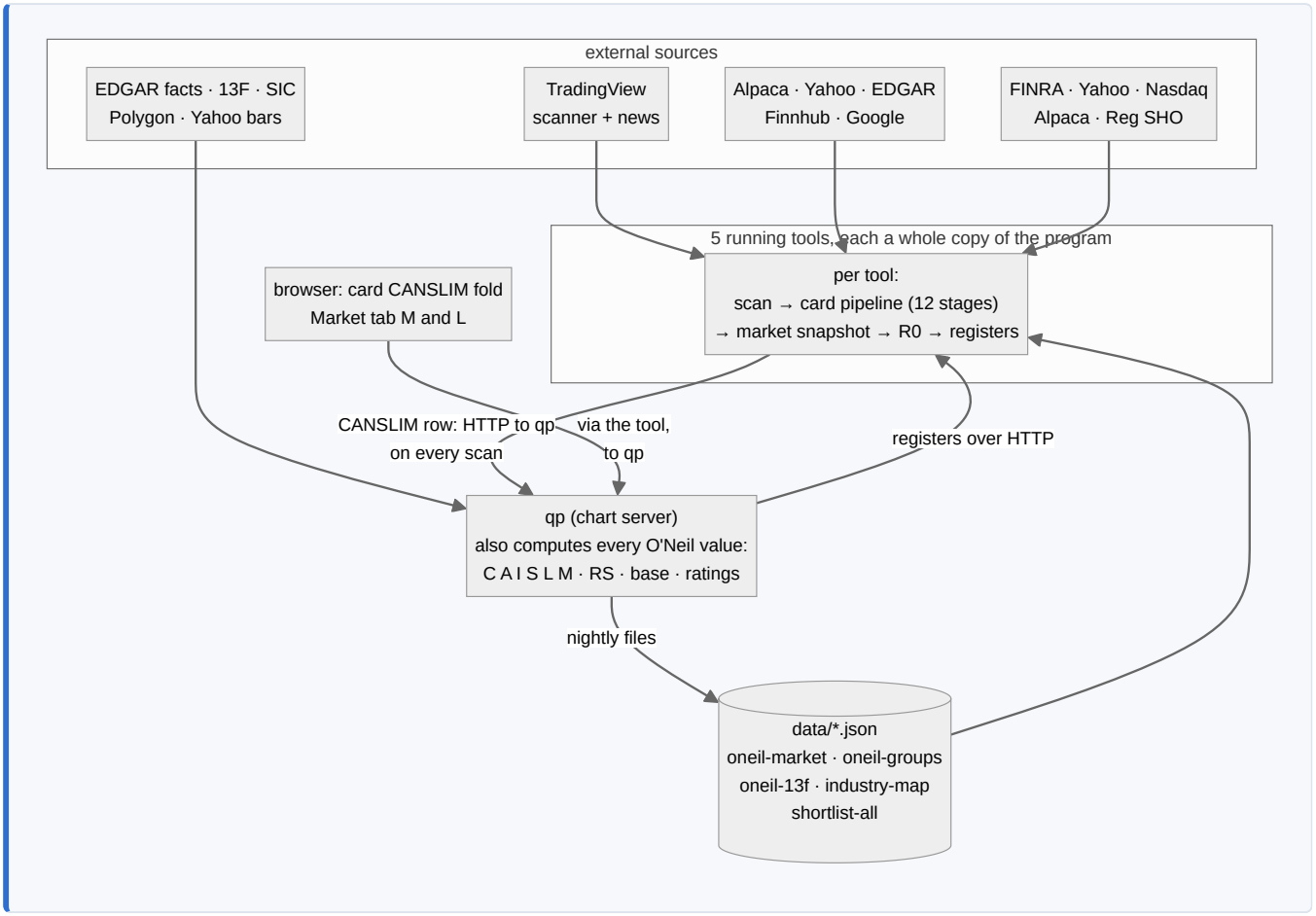
Where the data comes from

Ten external sources feed the desk today.

Source	Feeds	Cadence	Needs
TradingView scanner	candidates and their 32 columns; re-quotes every 5 min; the market snapshot (SPY, QQQ, IWM, DIA, VIX, 15 sector ETFs)	every scan	nothing
TradingView news	headlines	live cards, every scan	nothing
Yahoo Finance	news; short interest (fallback); daily bars of ^GSPC/^IXIC for the market model; outcome bars (R3); weekly bars for the base; chart bars	per scan, nightly, 16:05, on demand	nothing
Alpaca	Benzinga news; shortable flag; bars (move rate, R3 fallback, journal); account fills (journal)	per scan, daily, 16:05	key
FINRA	short interest (one file for every symbol)	daily	nothing
Nasdaq	short interest (fallback); Reg SHO threshold list	daily, per settlement day	nothing
NYSE	Reg SHO threshold list	per settlement day	nothing
SEC EDGAR	filings as news; company facts (C and A); SIC codes (industry groups); Form 13F (I)	per scan, nightly, quarterly	a User-Agent
Polygon	grouped-daily whole market (the RS rating); bars	nightly, on demand	key
Finnhub, **Google News**	news (Finnhub with a key; Google only when the others return fewer than 2 items)	live cards	key / nothing

Plus computed sources: derived numbers, relations, technical catalysts, themes, the desk regime, sector scores, RS ratings, group ranks, the base, up/down volume, the score.

Today, as it runs

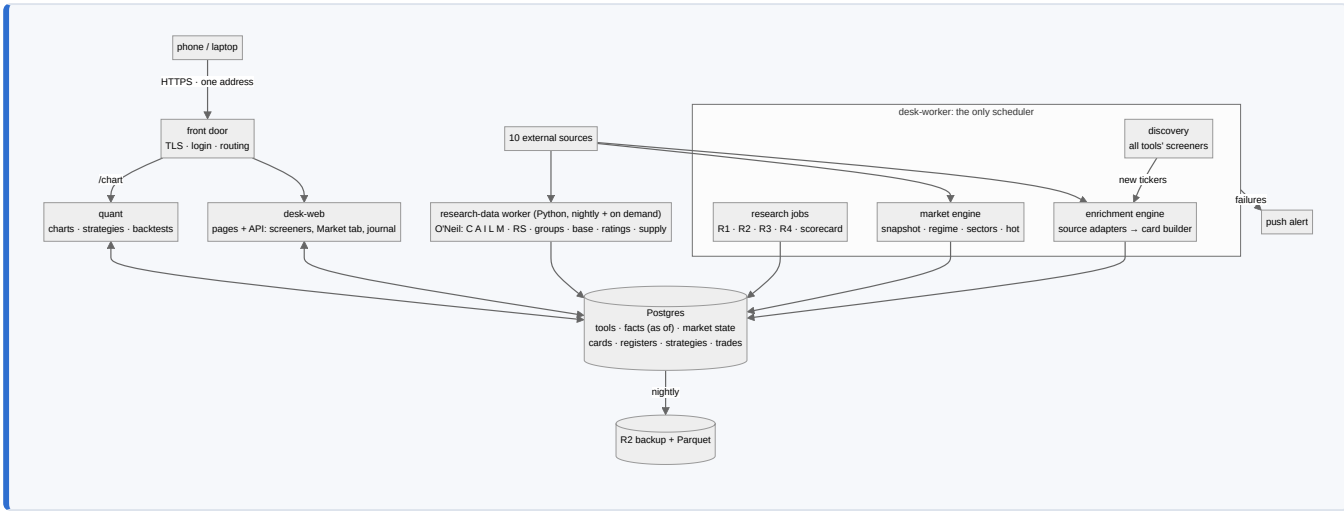


Found in the review	Root cause in the architecture
the same ticker’s news, short interest and borrow fetched by **every tool that found it**; five tools computing the **same market snapshot** from the same ETFs, minutes apart	the card pipeline and the market read live **inside each tool**, not as shared services
while qp restarts, every card’s CANSLIM fold is blank and every scan freezes its `cs*` columns as empty into R1; when an O’Neil file is missing, the Market tab asks the chart server to build it	**O’Neil values are computed inside the chart program**, though they are card fields and Market-tab sections
F-21: the industry map flipped between SIC and TradingView every morning	two writers for one shared file
facts with no time: short interest, borrow and news cached per process per day; the backtest has to reconstruct CANSLIM “as of” a day	**no fact store** with as-of timestamps
hot-sector state and live cards lost on every restart	state that matters kept in memory
S-1: anyone on the internet could change keys and strategies	thirteen public ports, no front door
D-S9: T6 photographs at 10:16 a screener that starts at 11:30	times and windows never checked against each other
qp restart took 3 min 20 s; the 113,854-restart incident	two supervisors in the past, port handover by guesswork
fixes verified only when someone runs the tests	no CI

The principles

Principle	Means	Prevents
The card is a product, built once	a ticker’s facts are fetched and computed once, by one enrichment engine, and every tool reads them	five tools fetching the same news and short interest; cards that disagree between tools
Every source behind an adapter	one adapter per source states what it returns, how fresh it must be, its fallback order and its honest null (“could not ask” is never “no”)	source logic scattered through the pipeline
Every fact carries when it became known	a fact store keyed by ticker, fact family and time	look-ahead (F-20, F-26, F-28); “as of” reconstructions
The market is read once	one market engine builds the snapshot, regime, sectors and O’Neil’s M and L for every tool, with history	five copies of the same read disagreeing
Research data is not the chart server’s job	the O’Neil chain runs as its own worker and writes to the fact store; the chart server only reads	a chart restart blanking cards
Isolate results, share facts	each tool’s candidates, registers, shortlist and model stay its own; facts about a ticker and the market are shared, as labels, never filters	the research rule
Tools are data	a tool is a row: its screeners, windows and capture times; one engine runs all of them	90 MB per tool; tools stopped for lack of memory
One writer for every fact	every table names its single owner; two sources for one fact are both stored, with a stated priority	F-21
One clock	one scheduler, with a run history and an alert when a run fails or does not happen	jobs spread over five processes, pm2 cron and systemd
Config checked when saved	capture before its screeners’ windows: refused, with the reason	D-S9
One front door	one HTTPS address, one login, no other open port	13 public ports
Proven before it ships	every pull request runs the tests; red blocks the merge	no CI

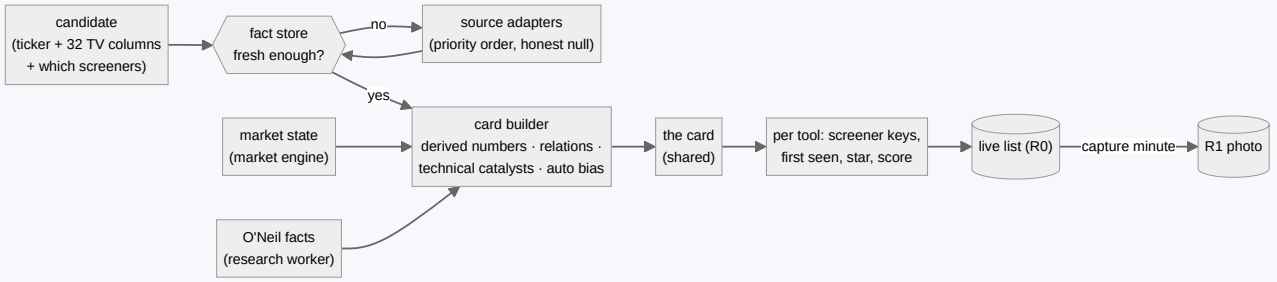
The target, in one picture



Part	Does
front door	HTTPS, the one login, routing; closes every other port
desk-web	every page: the screener suite, each tool’s tabs, the Market tab, the journal. Never runs a timed job
desk-worker	the only scheduler. Hosts discovery, the enrichment engine, the market engine and the research jobs; records every run; pushes an alert when one fails or does not happen
enrichment engine	the card builder (below)
market engine	the Market tab (below)
research-data worker	the O’Neil chain, moved out of the chart server (below)
quant	today’s qp: the verified library, charts, strategies, backtests, alerts. Reads registers and facts from the database, as of any date
Postgres	the system of record; on one small box, SQLite with one writer per file is the lighter alternative: the ownership rules matter more than the engine

The card, end to end

The enrichment engine turns a candidate into a card. It runs for every **new** ticker at once, and re-checks each fact family on that family’s own clock.


candidate
(ticker + 32 TV columns
+ which screeners)
fact store
fresh enough?
no
source adapters
(priority order, honest null)
yes
market state
(market engine)
card builder
derived numbers · relations ·
technical catalysts · auto bias
the card
(shared)
per tool: screener keys,
first seen, star, score
live list (R0)
capture minute
R1 photo
O'Neil facts
(research worker)

Fact family	Sources, in priority	Fresh for	On the card
quote, levels, ranges, volume, float, cap	TradingView scanner	5 minutes	header, Volume & Float, Range, Price, EMAs
derived numbers, relations	computed from the quote	with the quote	signal tags, PM/ADR, range positions
short interest, days to cover	FINRA → Yahoo → Nasdaq	a day	Volume & Float
borrow	Alpaca shortable; Reg SHO lists (Nasdaq, NYSE)	a day / settlement day	borrow note
news	TradingView, Alpaca, Yahoo, EDGAR, Finnhub; Google as fallback	each scan, live cards	News fold
catalyst, auto bias	computed from news (39 categories, 3 tiers) and the tape	with the news	badges, bias button
market context	the market engine	each scan	Market Context fold, ALL BULL/BEAR
industry, group	SIC (EDGAR) first; TradingView label only when SIC has none	nightly	header, L
C, A (earnings tables)	EDGAR company facts	nightly	CANSLIM fold
I (institutions)	Form 13F	nightly	CANSLIM fold
L (group rank, RS in group)	RS ratings (Polygon whole market) + groups	nightly	CANSLIM fold
M (market model per card)	the market model + the stock on distribution days	nightly + per scan	CANSLIM fold
base, U/D ratio, A/D, RS line	weekly and daily bars	nightly / on demand	CANSLIM fold
themes	ticker and industry lists	static	Themes
score	the tool’s own model	at discovery	score
shortlist marks	this tool; the union of every tool	live	☆, ★ also

Rules the engine keeps (each learned on the live desk):

- a missing value is **null with a reason**, never zero, never bearish (D-S2, F-14);
- a failed source never fails the card (soft stages);

- every fact is stored **with the time it became known**, so R1 freezes and backtests read exactly what was known then;
- shared facts are **labels, never filters**;
- one adapter per source; a second source for the same fact is stored beside the first with a stated priority (the F-21 lesson);
- each source's health is judged by its content, not by "it answered" (the data check).

The Market tab, end to end

The market engine builds the market state **once for every tool**, keeps its history, and stamps it on every card.

Section	Built from	When	Also used by
Indexes	TradingView: SPY, QQQ, DIA, IWM, VIX	each scan	—
Desk analysis: short-term bias, mid-term stage, long-term bias, final regime	TradingView index columns (SMAs, Bollinger, ADX, VWAP, RVOL)	each scan	every card's Market Context; the score; R2
Sector heatmap and **hot sectors**	15 sector ETFs; hot state counted **per trading day** and **persisted**	each scan	cards' 🔥 HOT, sector bias
M: O'Neil market model	^GSPC, ^IXIC daily bars (distribution days, rally attempts, follow-through)	nightly + on demand	cards' M; the backtest's CANSLIM gate
L: industry groups	SIC map + whole-market RS ratings	nightly	cards' L

Today the first three are computed separately by each tool on every scan, and the hot state is lost on restart; M and L are computed by qp.

O'Neil, placed correctly

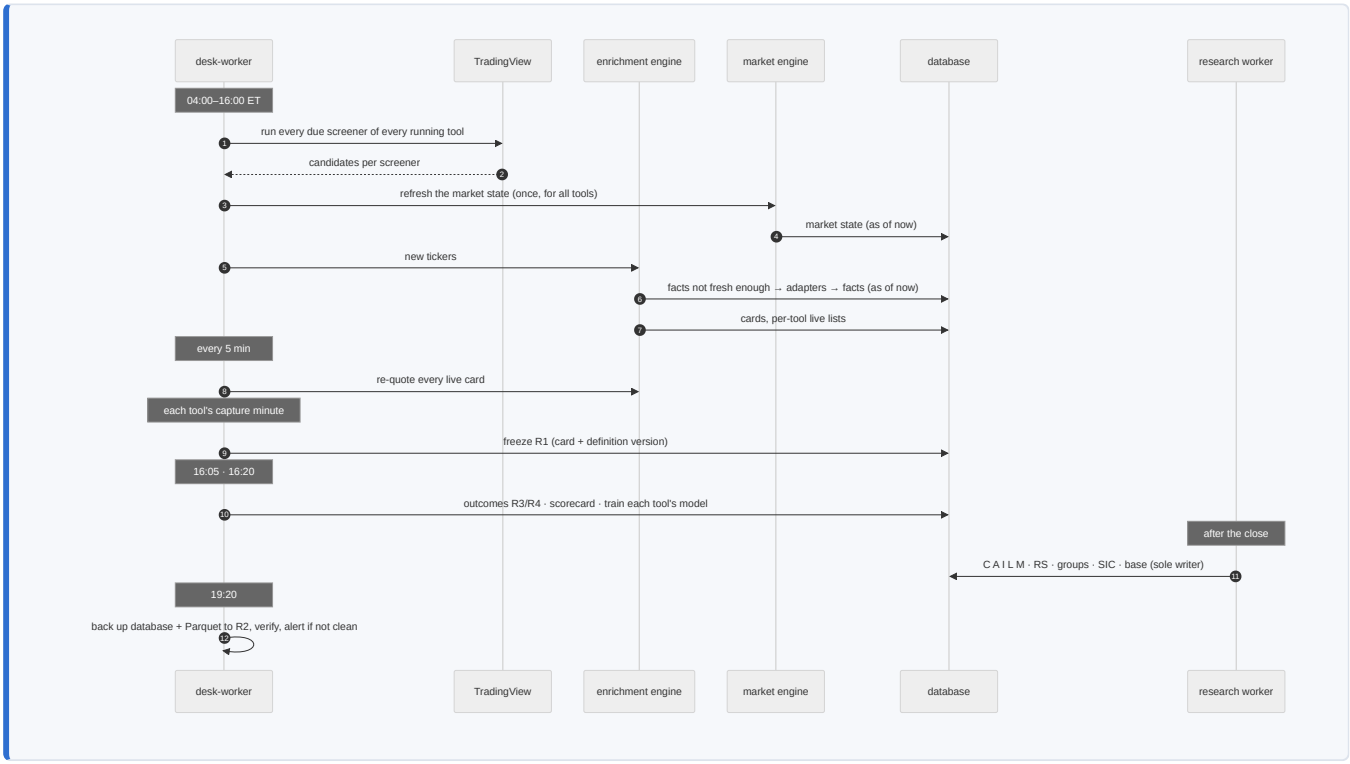
Today every O'Neil value is computed **inside qp**, because qp has the bars, pandas and the nightly job. But what O'Neil produces is **card fields** (the CANSLIM fold and the `cs*` columns frozen in R1) and **Market-tab sections** (M and L). That placement is why a chart restart leaves cards without their CANSLIM reading, and why every scan of every tool calls the chart server.

In the rebuild:

Piece	Owner	Read by
C, A (EDGAR), I (13F), supply (EDGAR shares), SIC map, RS ratings, groups (L), market model (M), base, ratings	**research-data worker**, writing facts with their dates	the enrichment engine (cards), the market engine (Market tab), quant (backtest gate, as of a day)
S and N on the card	the card builder, from those facts plus the card's own float, short interest, news and new highs	—
the CANSLIM fold, the `cs*` columns	the card	—
M and L sections	the Market tab	—

The method itself does not change (CANSLIM.md); only its home does.

A trading day, end to end



Where the data lives

Data	Owner (only writer)	Today
tools, screeners (versioned), windows, capture times	desk-web, validated on save	`tools.config.json` + `tools.local.json` + each tool's SQLite
ticker facts (quote, short interest, borrow, news, catalyst …), each with its time	enrichment engine	per-process caches and card JSON, per tool
cards (shared) and per-tool live lists	enrichment engine / desk-worker	each tool's memory, lost on restart
market state with history (snapshot, regime, sectors, hot state)	market engine	each tool's memory; hot state lost on restart
O'Neil facts (C, A, I, SIC, RS, groups, M, base, ratings)	research-data worker	`data/oneil-*.json`, `industry-map.json`, qp caches
registers R1–R4, shortlist	desk-worker	one SQLite per tool
strategies, backtests	quant	`~/.qp/platform.db`
trades	journal (in desk-web)	`data/journal/journal.db`
job runs	desk-worker	memory, per tool
secrets	you, in a secret store	files in `data/`, mode 0600

Memory, then and now

Program	Today	Target
screener tools (each with its own pipeline and market read)	5 × ~90 MB	inside desk-web + desk-worker
archive (6 stopped tools)	~80 MB	0 (same tables, read-only)
journal	~65 MB	inside desk-web
desk-web + desk-worker (one enrichment and market engine for all tools)	—	~150 + ~200 MB
quant	70–150 MB	70–150 MB
research-data worker	inside qp	runs as a job, then exits
Postgres	—	~60–100 MB
total, 5 tools	**~660–740 MB**	**~480–600 MB**
each extra tool	+~90 MB	a few MB: it adds screeners, not a pipeline

Today’s figures are from pm2 on the box (2026-10-06); target figures are estimates.

Getting there without a rewrite

Each step ships on its own, keeps the desk working, and can be undone.

#	Step	Risk
1	**CI on every pull request** (Jest and qp’s `run_all`)	none
2	**One front door** (Cloudflare Tunnel; close every port)	low · needs a domain
3	**Config checked on save** (capture vs windows)	low
4	**One writer per shared fact**: the `data/*.json` files become tables with owners	low–medium
5	**The fact store and the source adapters**: short interest, borrow and news move out of each tool into one shared service with per-family freshness; tools read from it. The first big saving: each ticker fetched once	medium · compare cards between old and new for a week
6	**The market engine, once**: one snapshot, regime and sector read for all tools; hot state persisted	low–medium
7	**The research-data worker**: the O’Neil chain leaves the chart server and writes facts; the card and Market tab stop calling qp	medium
8	**One scheduler with run history and alerts**	medium · run side by side for a week
9	**Tools as configuration**: one engine hosts every tool; R1 rows compared row by row before the old process stops	the largest step · reversible per tool
10	**Postgres and the Parquet lake**	medium · the backup proves the copy
11	**One web shell** (one bar, one origin)	low

What stays exactly as it is

- The research method: one photo a day per tool, entries A and B, outcomes in ATR, the benchmark, the mirrors.
- Every card field, every Market-tab section, every source and its fallback order: their **home** changes, their **content** does not.
- Each tool’s results, model and history isolated from every other tool.
- “A label, never a filter” for every shared fact.
- The O’Neil method as built, and qp’s verified library and engine.
- Archived tools stay readable; the nightly backup and the one password.

Decisions that are yours

- **Where it runs:** stay on EC2, or a managed host.
- **A domain name** for the one HTTPS address.
- **How far to go:** steps 1–7 remove most of the duplicated work and the risk found in the review; step 9 is where the per-tool memory cost goes away.